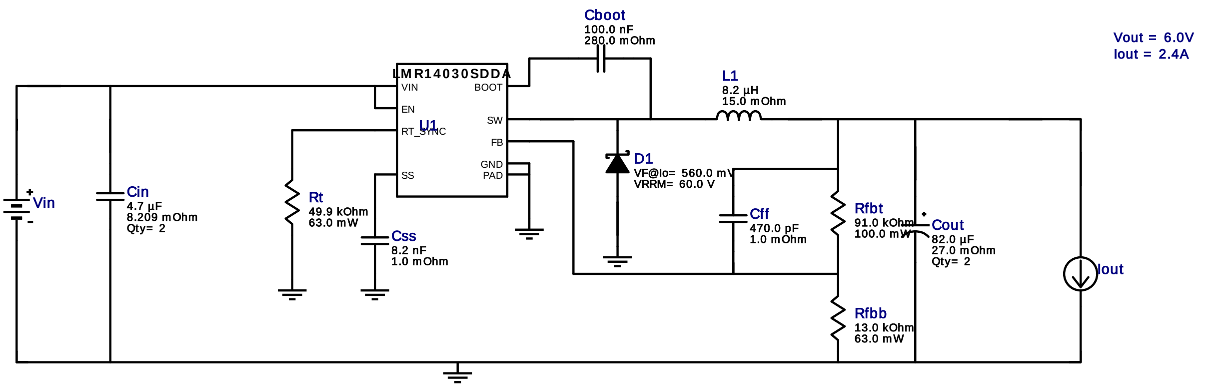
TEXAS INSTRUMENTS

VinMin = 14.8V
VinMax = 16.4V
Vout = 6.0V
Iout = 2.4A

Device = LMR14030SDDAR
Topology = Buck
Created = 2025-08-09 07:12:57.782
BOM Cost = $3.13
BOM Count = 13
Total Pd = 1.16W

WEBENCH ® Design Report

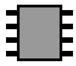
Design : 1 LMR14030SDDAR
LMR14030SDDAR 9V-12.6V to 6.00V @ 2.4A



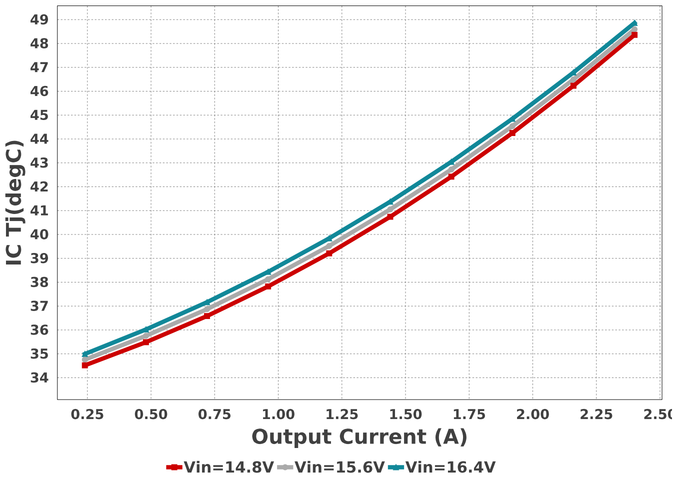
Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cboot	AVX	08053C104KAT2A Series= X7R	Cap= 100.0 nF ESR= 280.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	$0.01	0805 7 mm²
Cff	MuRata	GRM033R71C471KA01D Series= X7R	Cap= 470.0 pF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0201 2 mm²
Cin	Taiyo Yuden	MSAST21GAB5475MTNA01 Series= X5R	Cap= 4.7 uF ESR= 8.209 mOhm VDC= 25.0 V IRMS= 2.1621 A	2	$0.03	0805 7 mm²
Cout	Panasonic	16SVPF82M Series= SVPF	Cap= 82.0 uF ESR= 27.0 mOhm VDC= 16.0 V IRMS= 3.0 A	2	$0.47	CAPSMT_62_E61 53 mm²
Css	MuRata	GRM155R71C822KA01D Series= X7R	Cap= 8.2 nF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
D1	Diodes Inc.	PDS760-13	VF@Io= 560.0 mV VRRM= 60.0 V	1	$0.36	PowerDI5 50 mm²
L1	Bourns	SRU1048-8R2Y	L= 8.2 µH 15.0 mOhm	1	$0.50	SRU1048 144 mm²
Rfbb	Vishay-Dale	CRCW040213K0FKED Series= CRCW..e3	Res= 13.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
Rfbt	Yageo	RC0603FR-0791KL Series= ?	Res= 91.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	$0.01	0603 5 mm²
Rt	Vishay-Dale	CRCW040249K9FKED Series= CRCW..e3	Res= 49.9 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²

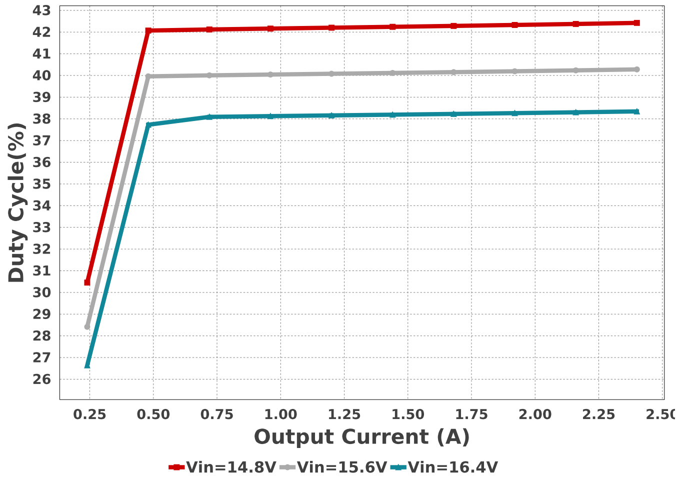
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
U1	Texas Instruments	LMR14030SDDAR	Switcher	1	$1.21	DDA0008E_N 55 mm^2



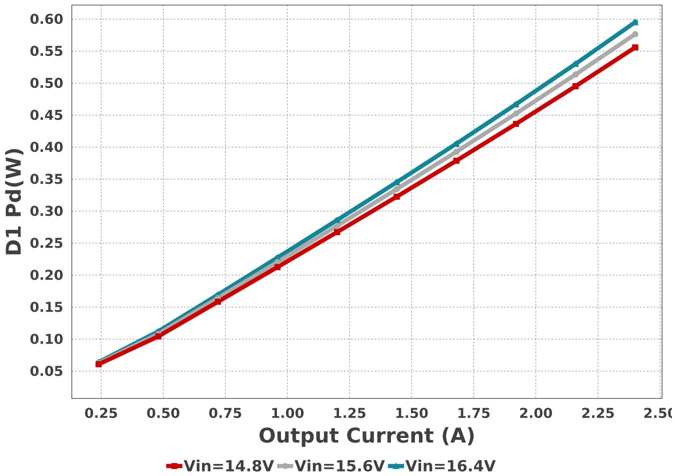
IC Tj



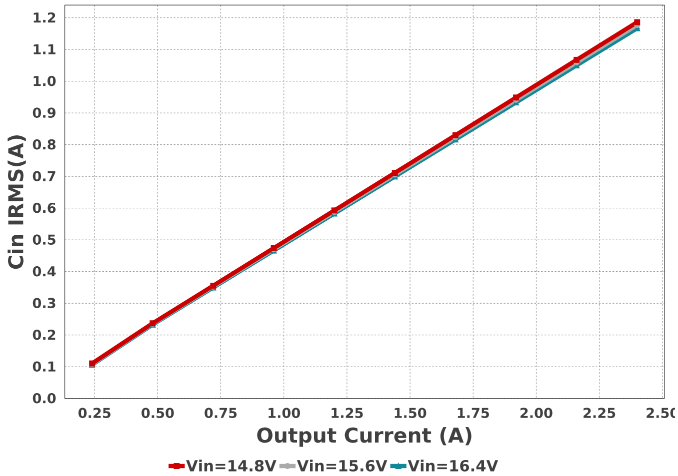
Duty Cycle



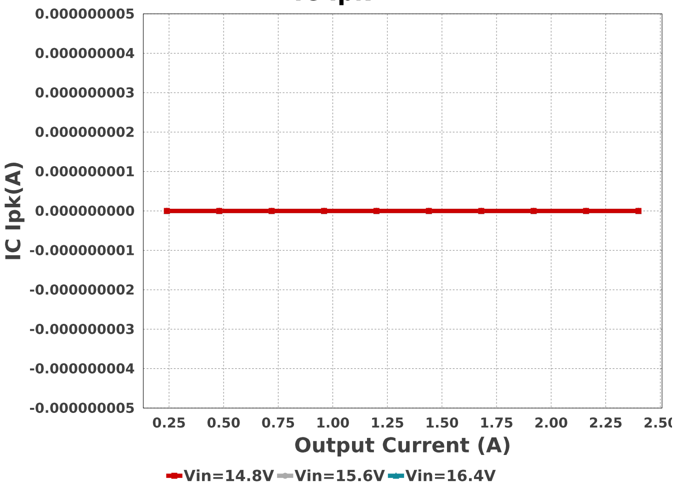
D1 Pd



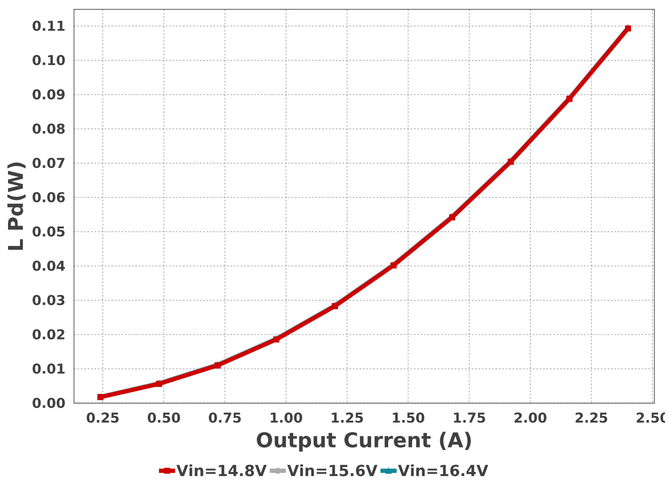
Cin IRMS

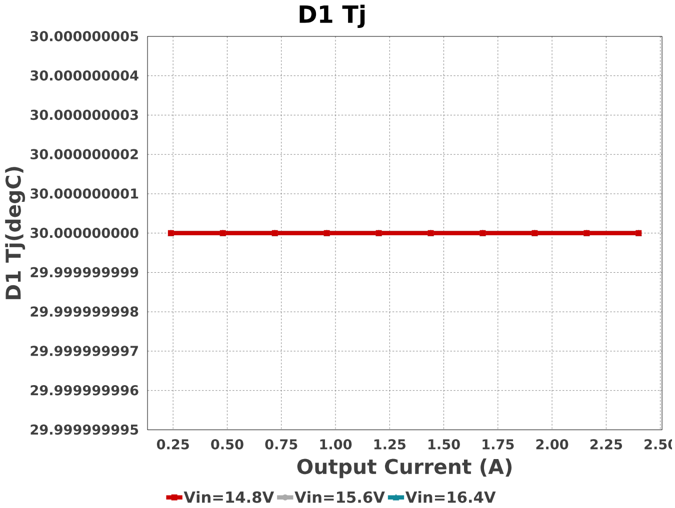


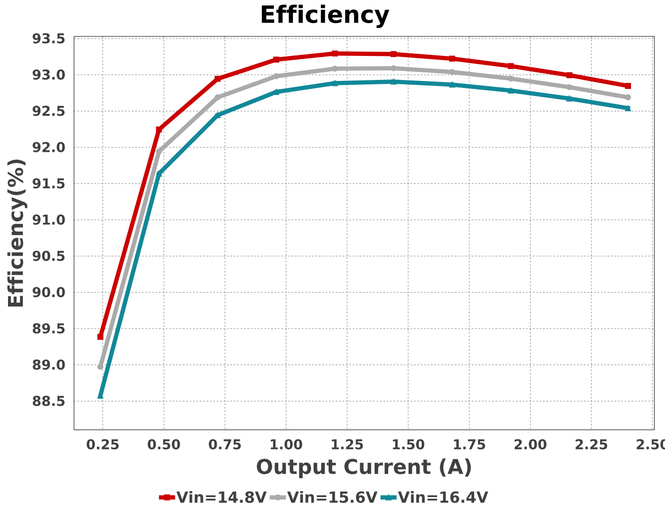
IC Ipk

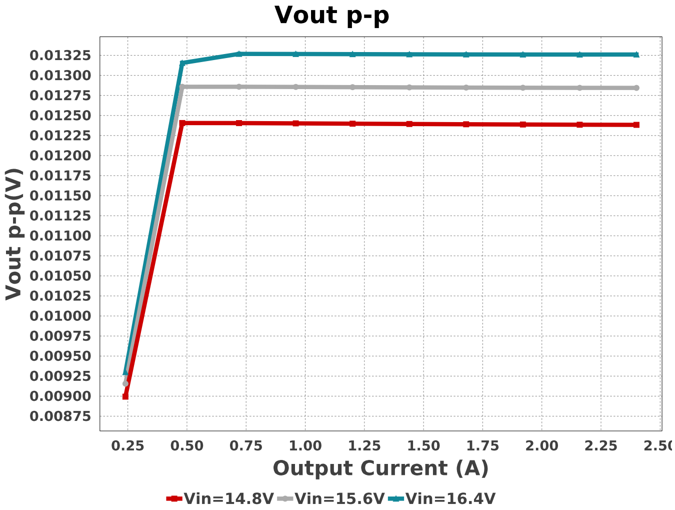


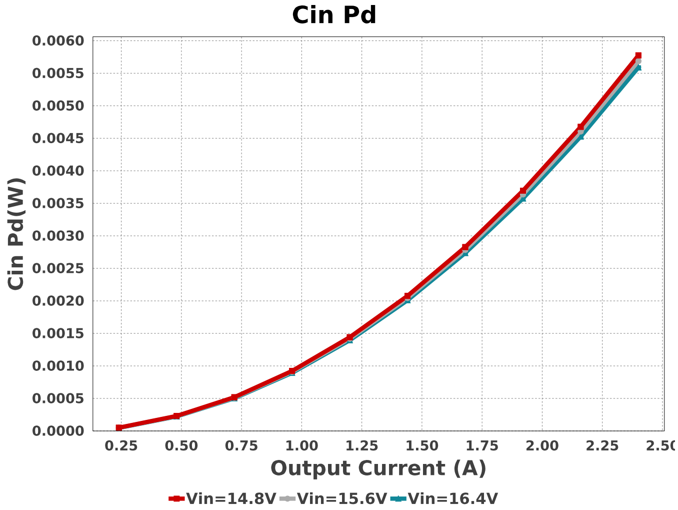
L Pd

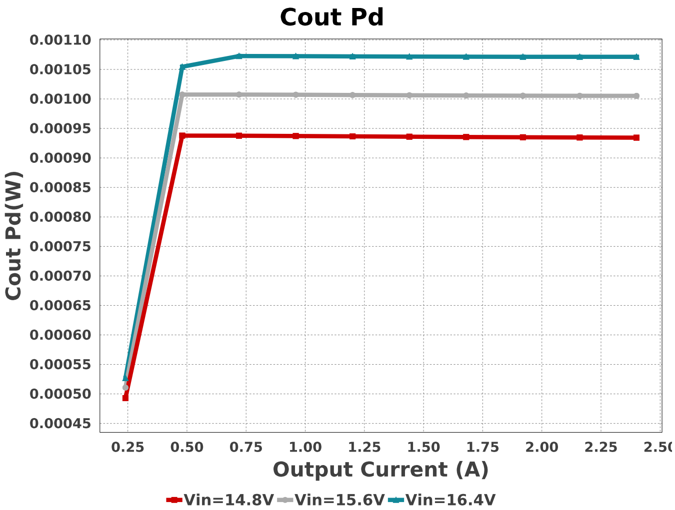


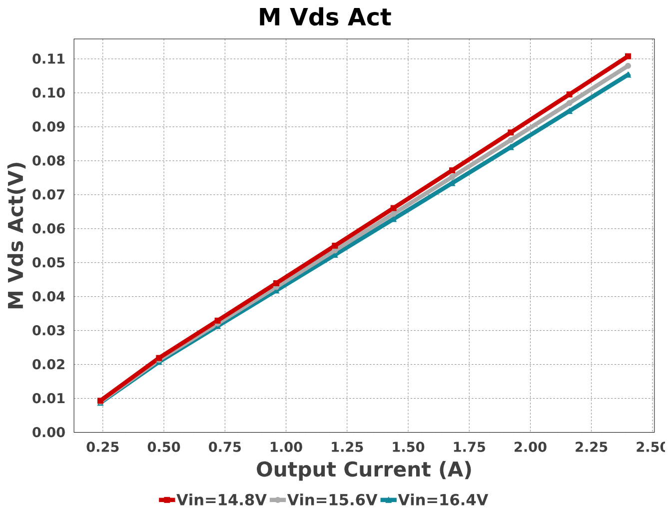

D1 Tj
D1 Tj(degC)
30.000000005
30.000000004
30.000000003
30.000000002
30.000000001
30.000000000
29.999999999
29.999999998
29.999999997
29.999999996
29.999999995
0.25 0.50 0.75 1.00 1.25 1.50 1.75 2.00 2.25 2.50
Output Current (A)
Vin=14.8V Vin=15.6V Vin=16.4V


Efficiency
Efficiency(%)
93.5
93.0
92.5
92.0
91.5
91.0
90.5
90.0
89.5
89.0
88.5
0.25 0.50 0.75 1.00 1.25 1.50 1.75 2.00 2.25 2.50
Output Current (A)
Vin=14.8V Vin=15.6V Vin=16.4V

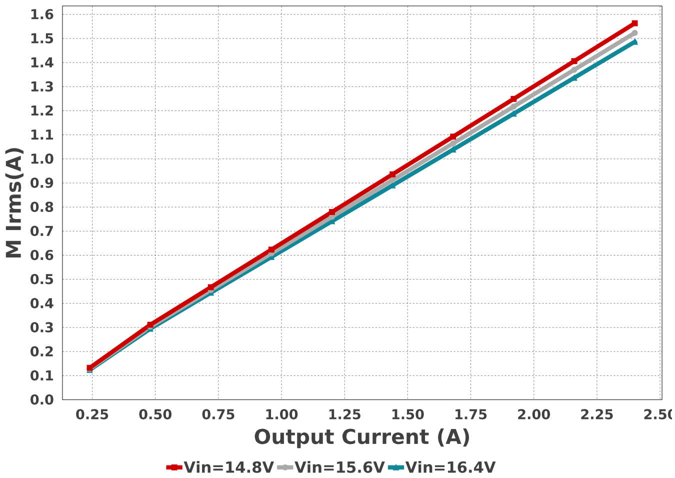

Vout p-p
Vout p-p(V)
0.01325
0.01300
0.01275
0.01250
0.01225
0.01200
0.01175
0.01150
0.01125
0.01100
0.01075
0.01050
0.01025
0.01000
0.00975
0.00950
0.00925
0.00900
0.00875
0.25 0.50 0.75 1.00 1.25 1.50 1.75 2.00 2.25 2.50
Output Current (A)
Vin=14.8V Vin=15.6V Vin=16.4V


Cin Pd
Cin Pd(W)
0.0060
0.0055
0.0050
0.0045
0.0040
0.0035
0.0030
0.0025
0.0020
0.0015
0.0010
0.0005
0.0000
0.25 0.50 0.75 1.00 1.25 1.50 1.75 2.00 2.25 2.50
Output Current (A)
Vin=14.8V Vin=15.6V Vin=16.4V

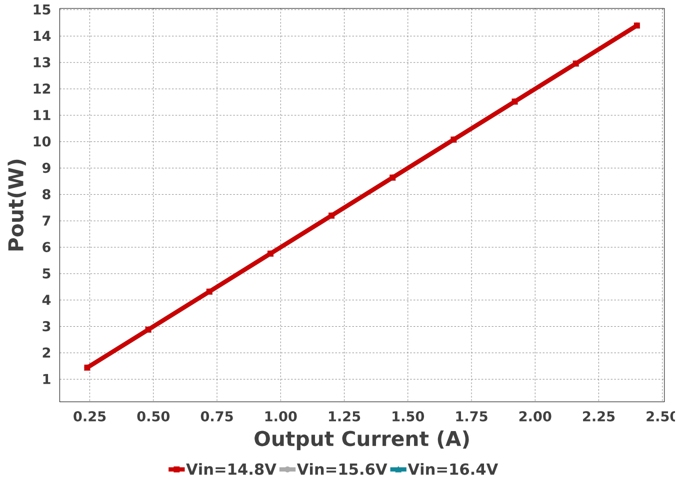

Cout Pd
Cout Pd(W)
0.00110
0.00105
0.00100
0.00095
0.00090
0.00085
0.00080
0.00075
0.00070
0.00065
0.00060
0.00055
0.00050
0.00045
0.25 0.50 0.75 1.00 1.25 1.50 1.75 2.00 2.25 2.50
Output Current (A)
Vin=14.8V Vin=15.6V Vin=16.4V


M Vds Act
M Vds Act(V)
0.11
0.10
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
0.25 0.50 0.75 1.00 1.25 1.50 1.75 2.00 2.25 2.50
Output Current (A)
Vin=14.8V Vin=15.6V Vin=16.4V

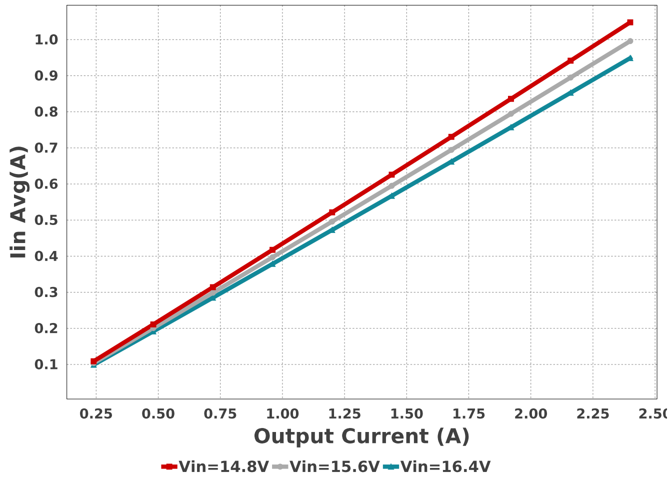
M Irms



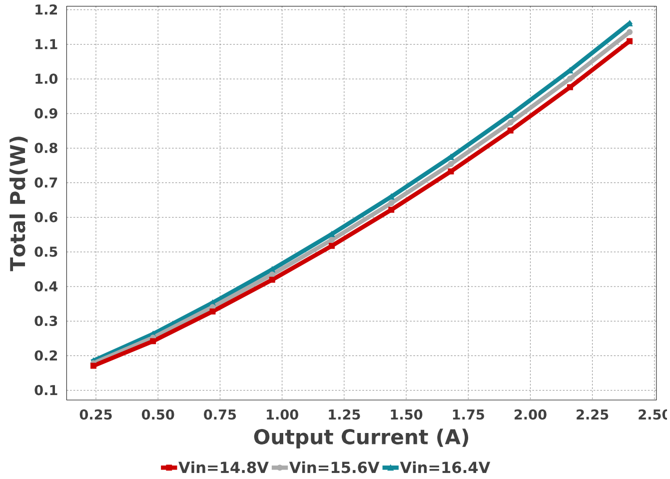
Pout



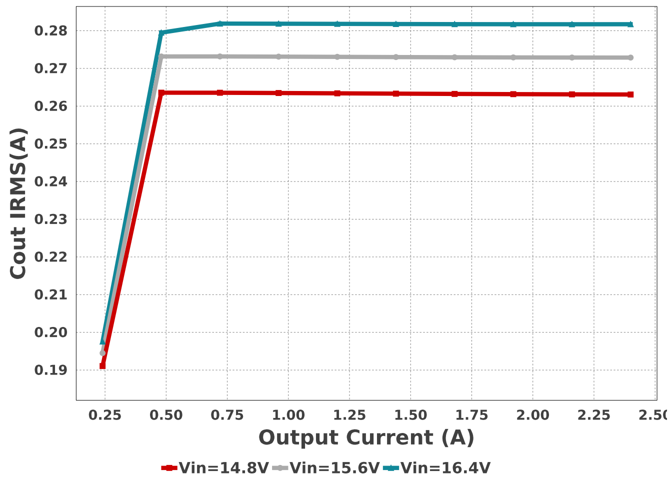
Iin Avg



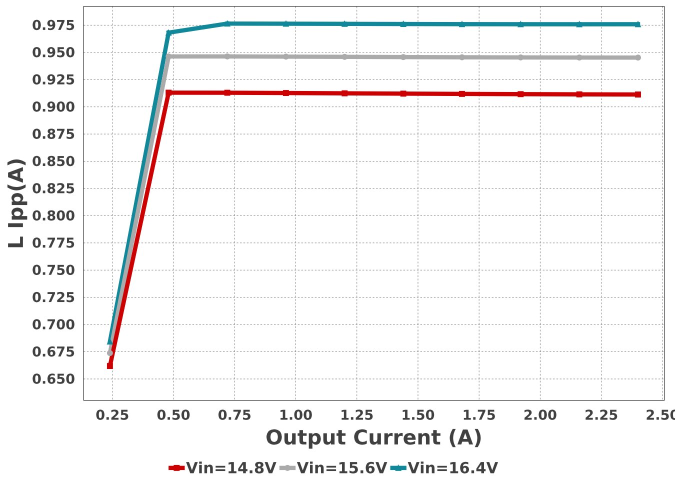
Total Pd

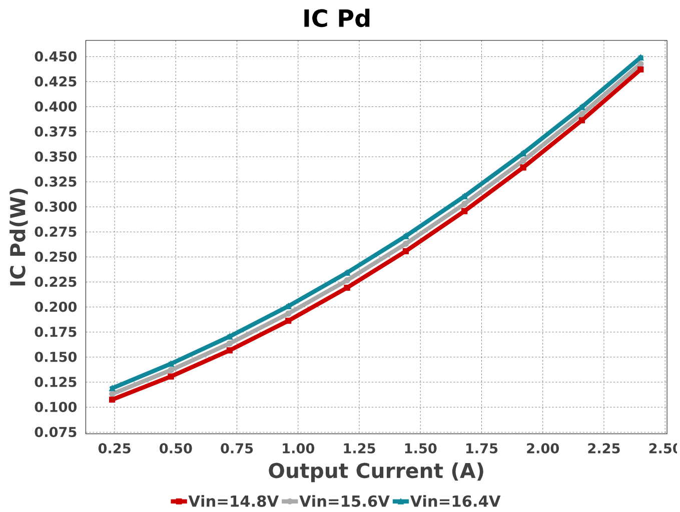


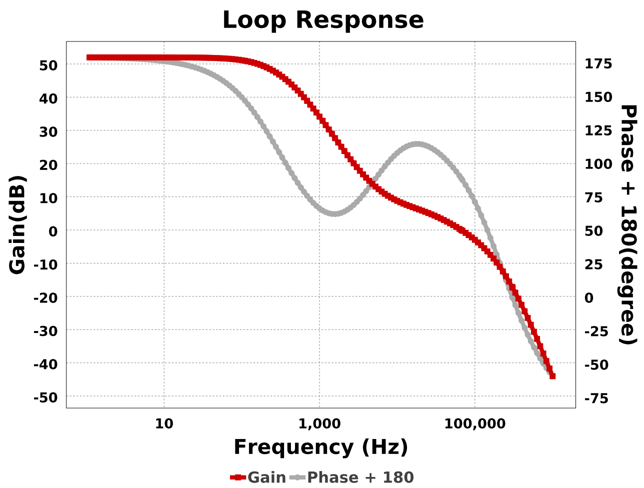
Cout IRMS



L Ipp







Operating Values

#	Name	Value	Category	Description
1.	BOM Count	13		Total Design BOM count
2.	Total BOM	$3.13		Total BOM Cost
3.	Cin IRMS	1.167 A	Capacitor	Input capacitor RMS ripple current
4.	Cin Pd	5.589 mW	Capacitor	Input capacitor power dissipation
5.	Cout IRMS	281.719 mA	Capacitor	Output capacitor RMS ripple current
6.	Cout Pd	1.071 mW	Capacitor	Output capacitor power dissipation
7.	D1 Pd	595.22 mW	Diode	Output Diode Power Dissipation
8.	D1 Tj	30.0 degC	Diode	D1 junction temperature
9.	IC Ipk	0.0 A	IC	Peak switch current in IC
10.	IC Pd	449.11 mW	IC	IC power dissipation
11.	IC Tj	48.862 degC	IC	IC junction temperature
12.	IC Tolerance	18.0 mV	IC	IC Feedback Tolerance
13.	ICThetaJA	42.0 degC/W	IC	IC junction-to-ambient thermal resistance
14.	Iin Avg	948.83 mA	IC	Average input current
15.	L Ipp	975.903 mA	Inductor	Peak-to-peak inductor ripple current
16.	L Pd	109.49 mW	Inductor	Inductor power dissipation
17.	M Irms	1.486 A	Mosfet	MOSFET RMS ripple current
18.	M Vds Act	105.342 mV	Mosfet	Voltage drop across the MosFET
19.	Cin Pd	5.589 mW	Power	Input capacitor power dissipation
20.	Cout Pd	1.071 mW	Power	Output capacitor power dissipation
21.	D1 Pd	595.22 mW	Power	Output Diode Power Dissipation
22.	IC Pd	449.11 mW	Power	IC power dissipation
23.	L Pd	109.49 mW	Power	Inductor power dissipation
24.	Total Pd	1.161 W	Power	Total Power Dissipation
25.	Cross Freq	66.653 kHz	System Information	Bode plot crossover frequency
26.	Duty Cycle	38.344 %	System Information	Duty cycle
27.	Efficiency	92.54 %	System Information	Steady state efficiency
28.	FootPrint	391.0 mm^2	System Information	Total Foot Print Area of BOM components
29.	Frequency	493.274 kHz	System Information	Switching frequency
30.	Gain Marg	-17.365 dB	System Information	Bode Plot Gain Margin
31.	Iout	2.4 A	System Information	Iout operating point
32.	Low Freq Gain	51.975 dB	System Information	Gain at 1Hz
33.	Mode	CCM	System Information	Conduction Mode
34.	Phase Marg	89.367 deg	System Information	Bode Plot Phase Margin
35.	Pout	14.4 W	System Information	Total output power
36.	Vin	16.4 V	System Information	Vin operating point
37.	Vout	6.0 V	System Information	Operational Output Voltage

#	Name	Value	Category	Description
38.	Vout Actual	6.0 V	System Information	Vout Actual calculated based on selected voltage divider resistors
39.	Vout Tolerance	4.21 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
40.	Vout p-p	13.261 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	2.4	Maximum Output Current
SoftStart	2.0 ms	Soft Start Time (ms)
VinMax	16.4	Maximum input voltage
VinMin	14.8	Minimum input voltage
Vout	6.0	Output Voltage
base_pn	LMR14030S	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature
UserFsw	500.0 k	Customer Selected Frequency

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of Cin and Cout, and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

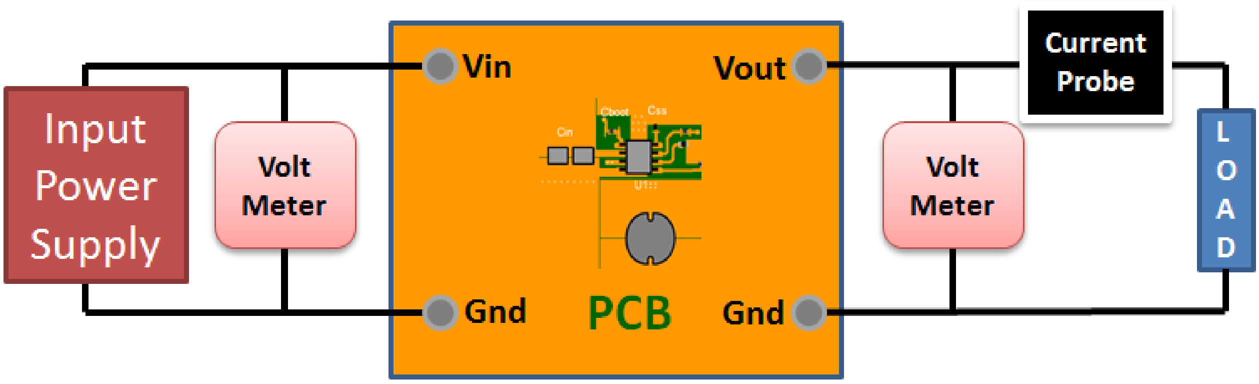
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 14.8V and set the input supply's current limit to zero. With the input supply off connect up the input supply to Vin and GND. Connect a digital volt meter and a load if needed to set the minimum Iout of the design from Vout and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between Vin and GND, a load is connected between Vout and GND and a current meter is connected in series between Vout and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.

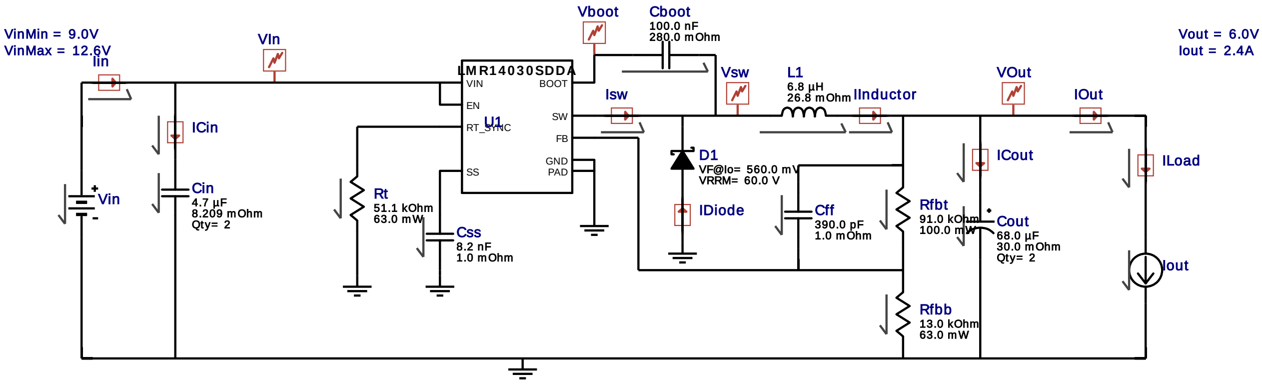


WEBENCH® Electrical Simulation Report

Design Id = 1

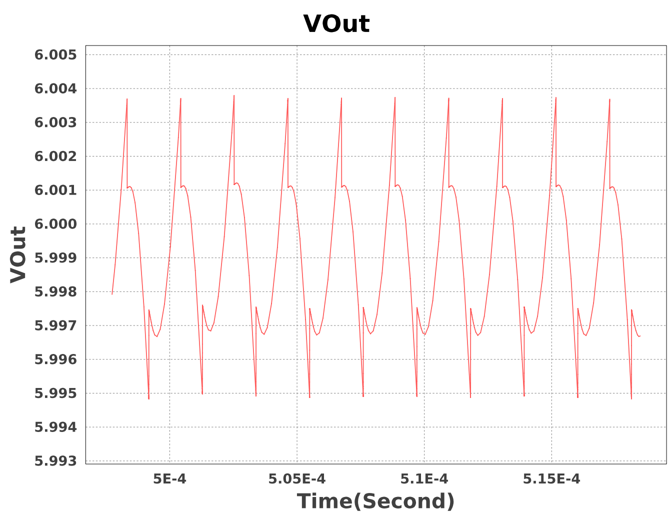
sim_id = 1

Simulation Type = Steady State



Simulation Parameters

#	Name	Parameter Name	Description	Values
1.	L1	IC	Initial Current	2.4 A
2.	Cboot	IC	Initial condition	10 V
3.	Iout	I	Load Current	2.4 A



Design Assistance

1. Master key : 39E66331F4444E9CA0EA904136A142C9[v1]

2. **LMR14030S** Product Folder : http://www.ti.com/product/LMR14030 : contains the data sheet and other resources.

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